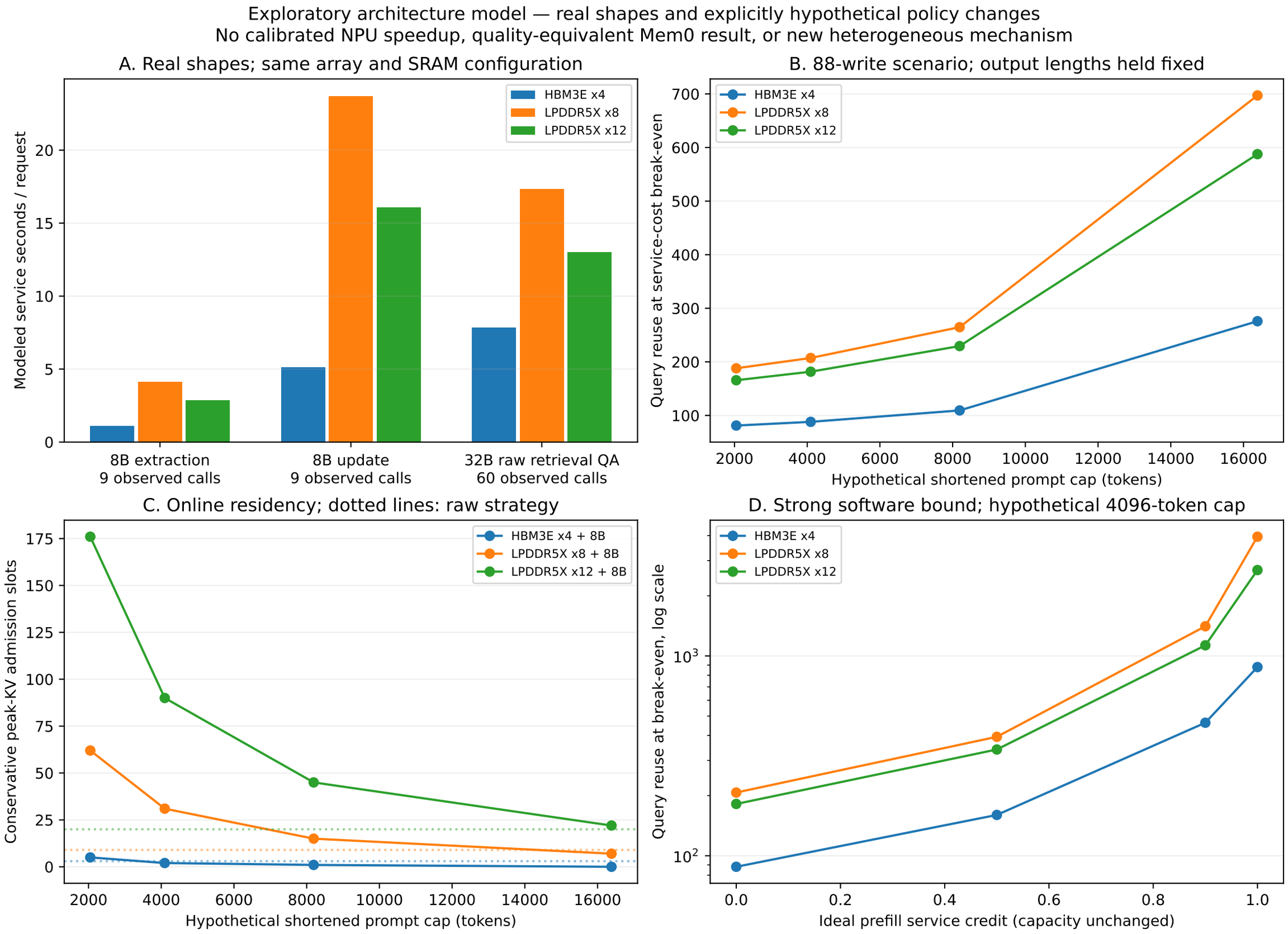
Exploratory architecture model — real shapes and explicitly hypothetical policy changes
No calibrated NPU speedup, quality-equivalent Mem0 result, or new heterogeneous mechanism
A. Real shapes; same array and SRAM configuration
Modeled service seconds / request
HBM3E x4
LPDDR5X x8
LPDDR5X x12
8B extraction
9 observed calls
8B update
9 observed calls
32B raw retrieval QA
60 observed calls
B. 88-write scenario; output lengths held fixed
Query reuse at service-cost break-even
Hypothetical shortened prompt cap (tokens)
C. Online residency; dotted lines: raw strategy
HBM3E x4 + 8B
LPDDR5X x8 + 8B
LPDDR5X x12 + 8B
Conservative peak-KV admission slots
Hypothetical shortened prompt cap (tokens)
D. Strong software bound; hypothetical 4096-token cap
Query reuse at break-even, log scale
Ideal prefill service credit (capacity unchanged)